

The interpolant residual as Loewner plus diagonal

$D_0 = W_Y - K_{YY}^{-1}$ has displacement rank 2; $\text{Bez}(P_Y, Q)$ does not recover D_0

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Abstract

After R431 the one negative square of the graph resolvent lives in the interpolant residual $D_0 = W_Y - S_0$ on the holes Y , not in an Euler factor of the source. This round asks whether D_0 is a Loewner/Bezout object, so that $\text{ind}_-(D_0)$ is zero-structure of a rational pair. Displacement rank $\text{rank}(YD_0 - D_0Y) = 2$ holds exactly over $\mathbb{Q}$ and with $\sigma_3/\sigma_2 \sim 10^{-14}$ on w_9 and two further windows. S_0 is exactly the inverse of the RKHS kernel K_{YY} of $P_{<d_0}$ on the X -measure. At kernel dimension 0, Cauchy- π additionally gives the dressed identity $\Delta D_0 \Delta = L_{-\tilde{m}} + \text{diag}(w_Y(P'_Y)^2)$ with $\tilde{m}$ the generating function of the $\pi_Y(X)^2$ -dressed X -measure. D_0 itself is *not* a classical ones-Loewner matrix once $|Y| \geq 3$ (the old trap: $n_Y = 2$ is vacuous). No source polynomial Q with P_Y -denominator makes $D_0 = S^T \text{Bez}(P_Y, Q) S$. The associated rational is m -like with poles at X , not at Y . Ausgang RESIDUAL_LOEWNER_PLUS_CONTROLLED_DIAGONAL. The new P1 problem is $\text{ind}_-(W_Y - K_{YY}^{-1}) \leq 1$. No RH claim.

Status. Lemmas 1–2 are proved (finite identities). Proposition 2 is a named miss. Propositions 1–3 are finite machine checks. Ausgang RESIDUAL_LOEWNER_PLUS_CONTROLLED_DIAGONAL. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The residual

Let X be the occupied dual nodes of a window, Y the holes, $u^\vee$ the r356 dual weights, and $d_0 = N_w - 3$ the interpolating depth of $\mathfrak{T}_0$ (r409). Write $W_Y = \text{diag}(u^\vee|_Y)$ and S_0 for the Gram of the min-norm interpolant in $P_{<d_0}$ (the Schur short-circuit of r409). The residual is

$$D_0 = W_Y - S_0, \tag{1}$$

so that $I - \mathfrak{T}_0^* \mathfrak{T}_0$ is the W_Y -congruence of D_0 . At kernel dimension $k_{\text{dim}} = d_0 - |Y| = 0$, R431 gives the Cauchy- π formula $S_0 = \Delta^{-1} C \Pi W_X \Pi C^T \Delta^{-1}$ with $\Delta = \text{diag}(P'_Y(y_i))$ and $C_{ia} = 1/(x_a - y_i)$. The one negative square of the bulk sits in D_0 .

2 Displacement rank two

Lemma 1 (Displacement). *Let $Y = \text{diag}(y_i)$. On the five-atom $k_{\text{dim}} = 0$ toy and on the six-node $\mathbb{Q}$ -window of the R431 audit, $\text{rank}(YD_0 - D_0Y) = 2$ exactly over $\mathbb{Q}$.*

A diagonal commutes with Y , so $YD_0 - D_0Y = -(YS_0 - S_0Y)$. At $k_{\text{dim}} = 0$, S_0 is a Δ -dressed Cauchy-Pick kernel, hence Cauchy-like of displacement rank 2. The identity $S_0 = K_{YY}^{-1}$ of Lemma 2 extends the same conclusion to every k_{dim} : the inverse of a Cauchy-like matrix is Cauchy-like (Gohberg-Semencul).

Proposition 1 (Windows). *On w_9 ($|Y| = 104$, $d_0 = 181$, $k_{\text{dim}} = 77$), $k_z = 15$ and $k_z = 18$, the singular values of $YD_0 - D_0Y$ are a repeated pair $(\sigma, \sigma, 0, 0, \dots)$. The ratio σ_3/σ_2 is $9.42 \cdot 10^{-15}$, $1.08 \cdot 10^{-14}$, $4.41 \cdot 10^{-15}$ (numpy); mpmath dps 40 on w_9 gives $9.42 \cdot 10^{-15}$ with $\sigma_3 = 1.30 \cdot 10^{-15}$. Λ -permute keeps $\sigma_3/\sigma_2 = 5.71 \cdot 10^{-14}$.*

The stop `DISPLACEMENT_RANK_GROWS` does not fire.

3 S_0 is the inverse CD kernel

Lemma 2 ($S_0 = K_{YY}^{-1}$). *Let K_{YY} be the reproducing kernel of $P_{<d_0}$ for the X -measure, evaluated at Y ($K = \Phi_Y G_X^{-1} \Phi_Y^T$, G_X the moment Gram). Then $S_0 = K_{YY}^{-1}$ exactly over $\mathbb{Q}$ on both toys, and to relative residual $1.05 \cdot 10^{-12}$ on w_9 . Consequently*

$$D_0 = W_Y - K_{YY}^{-1}. \quad (2)$$

This is the min-norm interpolation identity: the Gram of coordinate interpolants in value space is the inverse of the restricted kernel. K_{YY} is the Christoffel–Darboux kernel of the X -measure, a Bezout kernel of the orthogonal pair (p_{d_0}, p_{d_0-1}) , not of (P_Y, Q) .

4 The old trap, and the dressed Loewner

A classical Loewner matrix of one function f satisfies $(y_i - y_j)(L_f)_{ij} = f(y_i) - f(y_j)$ (ones-normalisation). On the five-atom toy $|Y| = 2$ this is vacuous (one pair): the off-diagonals determine a nodal f , the polynomial extension has L_f with $\text{ind}_- = 0$, and $\text{diag}(\Delta) = D_0 - \text{off}(L_f)$ carries $\text{ind}_- = 1$ — the wall-side trap of Rounds 184–191, now on D_0 . On the six-node window $|Y| = 3$ the ones-residual is $40073273/275164890 \neq 0$: D_0 is not $L_f + \text{diag}(\Delta)$ in the node basis.

Lemma 3 (Dressed Loewner at $k_{\text{dim}} = 0$). *Let $\tilde{m}(z) = \sum_a u^\vee(x_a) P_Y(x_a)^2 / (x_a - z)$ and $L_{\tilde{m}}$ its Loewner matrix at Y (diagonal $\tilde{m}'$). At $k_{\text{dim}} = 0$,*

$$\Delta D_0 \Delta = L_{-\tilde{m}} + \text{diag}(w_Y(P_Y')^2) \quad (3)$$

exactly over $\mathbb{Q}$. $L_{-\tilde{m}}$ is negative definite ($\text{ind}_- = |Y|$); the diagonal is positive definite.

The live pair is therefore $(L_{-\tilde{m}}, \text{diag}(w_Y(P_Y')^2))$, not a single-function Loewner of D_0 . The diagonal is source-pure (dual weights times $P_Y'^2$). It does not by itself carry $\text{ind}_-(D_0)$ (it is PD); $L_{-\tilde{m}}$ does not by itself carry it either (it is ND). The mixture is the P1 object. `DIAGONAL_SHIFT_IS_THE_TARGET` does not fire as a stop on this pair.

5 Bezout of (P_Y, Q) misses D_0

The contracted B1 asks for a source polynomial Q with $D_0 = S^T \text{Bez}(P_Y, Q) S$ and Q from the Birkhoff/interpolation formula, never from eigendata of D_0 . The Cauchy– π rational $\tilde{m}$ has poles at X , denominator P_X , not P_Y . The Lagrange interpolant of $\tilde{m}$ at Y is a polynomial Q of degree $< |Y|$ (source-first).

Proposition 2 (B1 miss). *For Q the interpolant of $\tilde{m}$ at Y , none of the source maps $S \in \{V^{-1}, \Delta^{-1}V^{-1}, V\}$ (V the Vandermonde at Y) satisfies $S^T \text{Bez}(P_Y, Q) S = D_0$ on either $\mathbb{Q}$ -toy. The last Sturm remainder of (P_Y, Q) on the five-atom toy is $-7466971/10681442 \neq 187/450 = 1 - q^\dagger$ ($r433$).*

The CD pair (p_{d_0}, p_{d_0-1}) represents K_{YY} , not D_0 . `RATIONAL_FUNCTION_USES_EIGENDATA` does not fire (Q is source). B1 does not stand, so B2 (Cauchy index of Q/P_Y) is not an identity for $\text{ind}_-(D_0)$. On w_9 , the residue count $\#\{i : \tilde{m}(y_i)/P'_Y(y_i) < 0\} = 52$ is not $\text{ind}_-(D_0) = 1$: that CInd is a $k_{\text{dim}} = 0$ count of holes, not P1 on the live window. `CAUCHY_INDEX_RESTATEMENT` is not claimed. The R437 edge contract (augmented $(P^\dagger, Q^\dagger)$ with last Sturm pivot $\delta = 1 - q^\dagger$) is not opened.

6 Worlds

Proposition 3 (Census). *MAIN $w_9/k_z15/k_z18$: $\text{ind}_-(D_0) = 1$. χ_3 -9: $\text{ind}_- = 0$ (PD, world-separating). χ_3 -15: $\text{ind}_- = 1$. Λ -permute / jitter / scramble send ind_- to 20/19/21. Displacement rank 2 survives the kills.*

In zero language the hoped census “MAIN/ χ at most one non-interlacing place, permute/scramble many” is a statement about $\text{ind}_-(W_Y - K_{YY}^{-1})$, not about interlacing of a P_Y -denominator Q .

7 The P1 problem

Remark 1 (Problem). Prove, source-purely, $\text{ind}_-(W_Y - K_{YY}^{-1}) \leq 1$ on the selected sequence (equivalently at $k_{\text{dim}} = 0$: the positive diagonal $w_Y(P'_Y)^2$ lifts all but at most one negative direction of $L_{-\tilde{m}}$). Sturm/Christoffel candidates: compare dual hole weights to the Christoffel function $1/K(y_i, y_i)$ of $P_{<d_0}$ on X , plus off-diagonal control of K^{-1} . Gauss-node interlacing of (p_{d_0}, p_{d_0-1}) makes K PD and does not by itself bound $\text{ind}_-(W_Y - K^{-1})$. This is a finite matrix problem per window. It is not a theorem of the Riemann hypothesis.

Mincut unchanged (base 4 / refined 5). No L^* claim. No $R^\dagger$ claim. No RH claim.